

Problems in Random Variables, Probability Mass and Density Function

1)

Find the formula for the probability distribution of the number of heads, when a fair coin is tossed 4 times. (MU — Nov. 96)

2)

A coin is known to come up heads 3 times as often as tails. This coin is tossed 3 times. Write down the probability distribution of the number of heads that appear and also the cdf. Make a sketch of both.

3)

Consider the experiment of tossing a fair coin 4 times. Define $X = 0$, if 0 or 1 head appears; $X = 1$, if 2 heads appear; $X = 2$, if 3 or 4 heads appear. Find the probability function, mean and variance of X .

4)

A discrete RV X has the following probability distribution.

x :	0	1	2	3	4	5	6	7	8
$p(x)$:	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

Find the value of a , $P(X < 3)$, variance and distribution function of X .

5)

The probability distribution of a RV X is given below:

x :	0	1	2	3
$p(x)$:	0.1	0.3	0.5	0.1

If $Y = X^2 + 2X$, find the probability distribution, mean and variance of Y .

6)

The probability mass function of a RV X is defined as

$P(X = 0) = 3C^2$, $P(X = 1) = 4C - 10C^2$ and $P(X = 2) = 5C - 1$, where $C > 0$ and $P(X = r) = 0$, if $r \neq 0, 1, 2$.

(i) Find the value of C , (ii) Find $P\{0 < X < 2/X > 0\}$, (iii) the distribution function of X , (iv) the largest value of X for which $F(x) < \frac{1}{2}$ and (v) the

smallest value of X for which $F(x) > \frac{1}{2}$.

7)

In a continuous distribution, the probability density is given by $f(x) = kx(2 - x)$, $0 < x < 2$. Find k , mean, variance and the distribution function.

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8)

The diameter of an electric cable X is a continuous RV with pdf $f(x) = kx(1 - x)$, $0 \leq x \leq 1$. Find (i) the value of k , (ii) cdf of X , (iii) the value of a such that $P(X < a) = 2P(X > a)$ and (iv) $P(X \leq 1/2 \mid 1/3 < X < 2/3)$.

9)

A continuous RV X that can assume values between $x = 2$ and $x = 5$ has a density function given by $f(x) = 2(1 + x)/27$. Find $P(3 < X < 4)$.

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10)

If the cdf of a continuous RV X is given by $F(x) = \frac{1}{2}e^{kx}$, $x \leq 0$, and $F(x) =$

$1 - \frac{1}{2}e^{-kx}$, $x > 0$, find $P(|x| \leq 1/k)$. Prove that the density function of X is

$$f(x) = \frac{k}{2} e^{-k|x|} - \infty < x < \infty, \text{ given that } k > 0.$$

11)

If the distribution function of a continuous RV X is given by $F(x) = 0$, when $x < 0$; $= x$, when $0 \leq x < 1$ and $= 1$, when $1 \leq x$, find the pdf of X . Also find $P(1/3 < X < 1/2)$ and $P(1/2 < X < 2)$ using the cdf of X .

12)

If a discrete probability distribution is given by $P(X = r) = k(1 - a)^{r-1}$, $0 < a < 1$, for $r = 1, 2, \dots, \infty$, find the value of k and also the mean and variance of X .

Answer:

1)

$$P(X = r) = 4 C_r \left(\frac{1}{2} \right)^4, r = 0, 1, 2, 3, 4$$

2)

$$P(X = 0) = \frac{1}{64}, P(X = 1) = \frac{9}{64}, P(X = 2) = \frac{27}{64}, P(X = 3) = \frac{27}{64}$$

3)

$$P(X = 0) = \frac{5}{16}, P(X = 1) = \frac{6}{16}, P(X = 2) = \frac{5}{16}; E(X) = 1; V(X) = \frac{5}{8}$$

4)

$$a = \frac{1}{18}; P(X < 3) = \frac{1}{9}; V(X) = 4.4719; F(x) = 0 \text{ in } x < 0, F(x) = \frac{1}{81} \text{ in } 0 \leq x < 1, F(x) = \frac{4}{81} \text{ in } 1 \leq x < 2, F(x) = \frac{9}{81} \text{ in } 2 \leq x < 3 \text{ etc. } F(x) = \frac{64}{81} \text{ in } 7 \leq x < 8 \text{ and } F(x) = 1 \text{ in } 8 \leq x.$$

5)

$$P(Y = 0) = 0.1; P(Y = 3) = 0.3; P(Y = 8) = 0.5; P(Y = 15) = 0.1; E(Y) = 6.4; V(Y) = 16.24.$$

6)

$$C = \frac{2}{7}; P = \frac{16}{37}; F(x) = 0, \text{ when } x < 0; = \frac{12}{49}, \text{ when } 0 \leq x < 1; = \frac{28}{49}, \text{ when } 1 \leq x < 2 \text{ and } = 1, \text{ when } 2 \leq x; x = 0; x = 1.$$

7)

$$k = \frac{3}{4}; E(X) = 1; V(X) = \frac{1}{5}; F(x) = 0, \text{ when } x < 0; = \frac{1}{4} (3x^2 - x^3), \text{ when } 0 \leq x < 2; = 1, \text{ when } 2 \leq x.$$

8)

- (i) $k = 6$;
- (ii) $F(x) = 0, \text{ when } x < 0; = 3x^2 - 2x^3, \text{ when } 0 \leq x < 1; = 1, \text{ when } 1 \leq x$;
- (iii) the root of the equation $6a^3 - 9a^2 + 2 = 0$ that lies between 0 and 1;
- (iv) $\frac{1}{2}$.

9)

$$\frac{1}{3}$$

10)

$$P = 1 - \frac{1}{e}; f(x) = \frac{k}{2} e^{-k|x|}$$

11)

$$f(x) = 1 \text{ in } 0 \leq x \leq 1 \text{ and } = 0, \text{ elsewhere; } \frac{1}{6}; \frac{1}{2}.$$

12)

$$k = 1; E(X) = e^t; V(X) = e^t (e^t - 1).$$